

Build-Readiness Audit: Can a 4-Person Non-Expert Team De-Novo-Design a Urethanase in 72 Hours?

TL;DR

- **Full de novo backbone-up enzyme design (Lauko-2025 / RFdiffusion2 style) targeting a urethanase is REACHABLE-BUT-RISKY for a non-expert team in 72 h**, but only if they (a) start from a *published* Ser-Sercis-Lys theozyme transplanted from UMG-SP2 (Paiva Chem. Sci. 2025, Świderek JACS 2025), (b) lean heavily on a hosted platform (Tamarind Bio or NeuroSnap) for at least the AlphaFold/RFdiffusion glue, and (c) treat "credible artifact" as a filter-cascade demo (hundreds-thousands of designs, ranked by AF2 pLDDT/scRMSD + PLACER preorganization), NOT a single "this enzyme works" claim.
- **The single biggest failure mode is not compute or licensing — it is the theozyme-specification step:** translating the published QM/MM transition state for the UMG-SP2 Ser-Sercis-Lys amidase into a correctly-formatted RFdiffusion2 atomic motif + Rosetta `(params)`/`(cst)` files. This is intellectual chemistry work, not engineering work, and it is what kills non-expert teams. De-risk it by literally copying the catalytic Ser190 / Ser166 / Lys91 side-chain coordinates and a bound short-chain carbamate out of the UMG-SP2 crystal structure (Li et al., *Adv. Sci.* 2025, DOI 10.1002/advs.202416019, 2.16–2.59 Å resolution) rather than generating a theozyme from scratch.
- **The realistic stretch artifact is:** ~500–2,000 RFdiffusion2 backbones scaffolding a UMG-SP2-derived Ser-Sercis-Lys + carbamate-TS motif → 4–8 LigandMPNN sequences each → Boltz-2 refold + binding-affinity → filter cascade (pLDDT > 75, global C α scRMSD < 1.5 Å, catalytic-C α RMSD < 1.0 Å — the exact thresholds Lauko et al. used in *Science* 2025 — plus PLACER apo catalytic H-bond formed in the great majority of 50 ensemble models) → top ~10–50 designs visualized in a React/FastAPI dashboard with FoldSeek novelty badges. Wet-lab catalytic validation is firmly out of scope and judges should be told so.

Key Findings

1. **Every required model is publicly available with permissive-enough licensing as of Dec 2025 / early 2026.** RFdiffusion (BSD), RFdiffusionAA (open), RFdiffusion2 (open since 15 Sept 2025, bioRxiv DOI 10.1101/2025.04.09.648075, accepted at *Nature Methods* 23, 96–105, Jan 2026, DOI 10.1038/s41592-025-02975-x**), RFdiffusion3 (open in Rosetta Commons Foundry since 22 Dec 2025**), bioRxiv DOI 10.1101/2025.09.18.676967), ProteinMPNN/LigandMPNN (MIT), Boltz-1/Boltz-2 (MIT, code+weights, both academic and commercial use OK), PLACER (open with weights, PNAS DOI 10.1073/pnas.2427161122), ColabFold (MIT), and Rosetta/PyRosetta (free for academic, paid for commercial). The two notable restrictions are **AlphaFold 3** (code CC-BY-NC-SA 4.0;

weights gated, request-only, non-commercial, classified by Google as Confidential Information per its WEIGHTS_TERMS_OF_USE.md) and **ESM3** (Cambrian non-commercial license; ESM3-open-small 1.4B is the only weights variant downloadable, and only for non-commercial use; ESM3 medium/large are API-only via EvolutionaryScale Forge). (Genengnews)

- The hackathon pipeline is well-documented and reproducible.** Open community wrappers (lauramarie99/ProtDesign2, laukoag/serine-hydrolase-design), the Meiler Lab RFdiffusionAA tutorial PDF) provide working diffuse → LigandMPNN → ColabFold → filter scripts you can clone and adapt. The Baker (laukoag/serine-hydrolase-design) GitHub repo directly contains a Ser-His + tetrahedral-substrate "stub" sampler (invrotzyme.py) plus Rosetta enzyme-constraint files — the closest existing template to what you would need for a urethanase.
- Compute is feasible on a single 24 GB GPU**, but slow. RFdiffusion2's README says "The demo generates 150 residue proteins with many of the residues atomized, so it can take upwards of 30 minutes to run all cases. Each case may take up to 10 minutes on an RTX2060." On an A100, the published RFdiffusion-v1 NVIDIA NIM benchmark is ~507 aa·step⁻¹·s⁻¹, implying ~15 s per 150-residue backbone at 50 inference steps; so an A100 can plausibly produce hundreds of backbones per hour. **Boltz-2 runs in ~20 s per structure on a single A100** (MIT/Recursion press release, June 2025: "In just 20 seconds, Boltz-2 reaches the performance of FEP that usually takes from 6 to 12 hours" — Saro Passaro), and **~1-5 min per ≤1300-token system on an RTX 4090** with bfloat16 (UCSF ChimeraX Boltz testing, April 2025: 9fz5 at 1025 tokens in 3 min, 8sa0 at 1371 tokens in 5 min). PLACER takes 1-3 s/model on GPU per its README (~1 min/50-model ensemble per design). (GitHub + 4)
- Tamarind Bio is the cleanest no-DevOps path** for a non-expert team. It exposes RFdiffusion, RFdiffusion2, RFdiffusion3, RFdiffusionAA, ProteinMPNN, LigandMPNN, SolubleMPNN, RFantibody, RoseTTAFold All-Atom, ProteinMPNN-ddG, FoldSeek, Boltz-1x, and a chained "Protein Design" pipeline (RFdiffusion → MPNN → AlphaFold filter), all behind a single web UI and a Python API. Its **free tier is exactly 10 monthly jobs** with unlimited storage and email support (per tamarind.bio/pricing), with an academic-upgrade form available for university/non-profit emails. Budget-wise this means using Tamarind for the demo top-K designs and running the high-throughput sampling locally or on rented A100s. (Y Combinator + 2)
- The Baker lab's own published filter cascade is the right design-metric stack — and the exact thresholds are public.** In Lauko et al., *Science* 2025, "AF2 was run using model 4 with three recycles. Designs were filtered for a global Cα RMSD < 1.5 Å, pLDDT > 75, and catalytic residue Cα RMSD < 1.0 Å." Downstream, PLACER ensemble preorganization (frequency of catalytic H-bond formation across 50 stochastic models, apo state) cut 98.4% of AF2-passing designs but raised wet-lab hit-rates from 1.6% → 5.2%, and a two-state (apo + acyl-enzyme intermediate) filter raised it further. For RFdiffusion2, the in-silico success criterion (Ahern et al. 2025) is: catalytic-heavy-atom RMSD < 1.5 Å to a Chai-1 prediction for

at least one LigandMPNN sequence, with no protein-ligand atoms within 1.5 Å (clash check). Hitting these thresholds in silico is the credible hackathon artifact even without wet-lab. [bakerlab](#) [nature](#)

Details

PART A — The de novo enzyme design workflow, explained from scratch

Modern de novo enzyme design is a 6-stage pipeline. Each stage takes a specific file format in, produces another file format out, and the stages chain.

A.1 Theozyme definition (the hard intellectual step). A "theozyme" is a quantum-chemistry-informed description of the *minimum constellation of catalytic functional groups* arranged around the transition state of the reaction you want to catalyze. For a serine hydrolase / amidase, that means: a nucleophilic Ser side-chain oxygen positioned ~3 Å from the substrate carbonyl carbon at the proper attack angle (the Bürgi-Dunitz trajectory), a base (His, Lys, or — in amidase-signature enzymes — another Ser activated by Lys) to deprotonate that Ser, and an "oxyanion hole" of two backbone NH groups or hydroxyls that stabilize the negatively-charged tetrahedral intermediate. Input format: a small PDB containing only the catalytic side-chain atoms and the bound substrate/transition-state mimic; plus a Rosetta [cst](#) constraint file (or [params](#) file) specifying bond/angle tolerances and which atoms are matched. Output: a 5-15-atom motif PDB that downstream tools consume. [nature](#) [bakerlab](#)

For a urethanase, the published QM/MM mechanism is exceptionally well characterized:

- **Paiva et al., *Chem. Sci.* 2025, 16, 2437–2452, DOI 10.1039/D4SC06688J:** hybrid QM/MM on UMG-SP2 cleaving DUE-MDA (a diurethane PU substrate). Establishes the Ser190nuc-Ser166cis-Lys91 triad and reports the rate-limiting acylation transition state at $\Delta G^\ddagger = 20.8$ kcal/mol. Reports the catalytic-residue protonation states and identifies stabilizing point mutations.
- **Świderek, Arafet, de Sousa Batista et al., *J. Am. Chem. Soc.* 2025, 147(46), 42511–42523, DOI 10.1021/jacs.5c13147:** independent QM/MM on UMG-SP2 with the model substrate 4-nitrophenyl benzylcarbamate (pNC). Reports a three-step esterase-like mechanism (acylation, hydrolysis, decarboxylation) with constant-pH nMD/MC titration of the active-site Lys. Used an AlphaFold2 model "prior to the availability of the crystal structure" — directly relevant evidence that the AF2-then-modify workflow works for this family.
- **Li, Han, Cong et al., *Adv. Sci.* 2025, 2416019, DOI 10.1002/advs.202416019:** published the UMG-SP2 crystal structures at 2.16/2.40/2.59 Å (ligand-free, suicide-inhibitor-bound, short-chain-carbamate-bound). Identifies the regulatory L3 loop (residues 219–226). [ADS](#)

A.2 Backbone scaffolding (RFdiffusion / RFdiffusion2 / RFdiffusion3 / RFdiffusionAA). Given a theozyme motif, generate a protein backbone that can hold those catalytic residues in that exact 3D arrangement. RFdiffusion (original, 2023) does this at the residue level — you must enumerate side-chain rotamers and place the backbone N-C α -C atoms. RFdiffusion-All-Atom (RFAA, 2024) adds atom-level ligand awareness. **RFdiffusion2** (Ahern et al., open-sourced 15 Sept 2025) is the breakthrough relevant to enzyme design: it scaffolds **directly from atomic functional-group coordinates** without requiring you to pre-enumerate rotamers, *and* it does not require you to specify the sequence index of each catalytic residue. The model successfully scaffolds 41/41 sites in the AME benchmark vs 16/41 for prior tools (Nature Methods Fig. 3c). In the experimental cases, "fewer than 96 sequences" were tested per catalytic motif: retroaldolase (4 active variants, best k_{cat}/K_M 6.34 M⁻¹s⁻¹); cysteine hydrolase (48 tested, best k_{cat}/K_M 248 M⁻¹s⁻¹); zinc-metallohydrolase (96 tested with a general base added, 11 functional, best k_{cat}/K_M 53,000 M⁻¹s⁻¹). **RFdiffusion3** (Butcher, Krishna et al., open-sourced 22 Dec 2025 via Rosetta Commons Foundry) is a unified atom-level diffusion model $\sim 10\times$ faster than RFdiffusion2, with cysteine hydrolase enzymes tested in 190 designs yielding "35 of 190 designs (18%) qualified as multi-turnover hits" with the best at $k_{cat}/K_M = 3.6 \times 10^3$ M⁻¹s⁻¹. Input: theozyme motif PDB + a "contig" string specifying total length (e.g. `150-200`) + optional ligand. Output: PDB backbone with poly-glycine residues but correct geometry, plus a `.trb` metadata file mapping motif residues to scaffold residues. (Society + 10)

A.3 Sequence design (ProteinMPNN / LigandMPNN / SolubleMPNN). Given a backbone PDB with empty (poly-glycine) residues, assign amino-acid identities that will fold to that backbone. ProteinMPNN (Dauparas et al., *Science* 2022) is sequence-only; **LigandMPNN** (Dauparas et al., MIT licensed, `dauparas/LigandMPNN`) extends it to be aware of bound ligands and is the right choice for enzyme design because it sees the substrate during sequence assignment. SolubleMPNN is a fine-tune that biases toward soluble surfaces. Input: PDB + a residues-to-fix list (typically the catalytic Ser/Ser/Lys) + optional ligand atoms. Output: a FASTA of 4-32 designed sequences per backbone plus an updated PDB with packed side-chains. Runs comfortably on a single 24 GB consumer GPU; sub-second per sequence.

A.4 Structure prediction self-consistency check (AF2/AF3/ESMFold/Boltz/Chai). Take each designed sequence and predict its structure from scratch. Compare to the originally-diffused backbone. If they match (low scRMSD), the design is "designable." The community-standard tool is **ColabFold** (Mirdita et al., *Nat. Methods* 2022) running AlphaFold2 with three recycles and reduced MSA — sub-minute per 200-residue sequence on an A100. For enzyme design where you also want to predict the ligand pose, **Boltz-2** (Passaro et al., bioRxiv DOI 10.1101/2025.06.14.659707, MIT, `jwohlwend/boltz`) is the current open-source state of the art: it handles protein + small-molecule ligand + binding affinity in ~ 20 s/structure on an A100, achieves Pearson $r \approx 0.62$ on the FEP+ benchmark (CDK2/TYK2/JNK1/P38) versus $r \approx 0.72$ for full FEP+, at $>1000\times$ the speed (Passaro et al. 2025). It was explicitly used by Ahern et al. as an alternative to AF3 (which is not fully open and is non-commercial). **Chai-1** (`chaidiscovery/chai-lab`) is also available under a permissive license, runs on a 4090, and is

what RFdiffusion2 uses internally for its in-silico success filter. **Chai-2 is API-only** and is specifically targeted at antibody design ("zero-shot antibody design in a 24-well plate"); not relevant to enzyme design. (BioLM + 5)

A.5 Active-site preorganization (PLACER + Rosetta). Structure prediction alone tells you the global fold is right, not that the catalytic geometry is actually formed. **PLACER** (Anishchenko, Kipnis, Kalvet et al., *PNAS* 122, e2427161122, 2025; (baker-laboratory/PLACER)) is a graph neural network that, given a protein structure plus a ligand, stochastically generates 50–200 plausible local conformational ensembles of the active site. For each ensemble you compute the **fraction of models in which key catalytic hydrogen bonds are correctly formed** — this is the "preorganization" metric. In Lauko et al. *Science* 2025, gating designs on apo-state Ser-His H-bond formation cut 98.4% of AF2-passing designs but raised the wet-lab hit-rate from 1.6% → 5.2%; a multi-state filter (apo + acyl-enzyme intermediate) raised the FP-probe labelling rate to 84% and 4MU-substrate hydrolysis rate to 18% across 132 round-3 designs. PLACER runtime per its README: "One model of an average ligand-protein complex will be predicted on a GPU within 1-3 seconds." A standard 50-model ensemble per state is therefore ~1 min/state; a 5-state ensemble (apo, substrate-bound, TI1, AEI, TI2) is ~5–15 min/design. (PNAS + 4)

A.6 Filtering / ranking. The published recipe is a cascading filter (Lauko et al. *Science* 2025, methods): "Designs were filtered for a global C α RMSD < 1.5 Å, pLDDT > 75, and catalytic residue C α RMSD < 1.0 Å" — then PLACER preorganization on the survivors. For RFdiffusion2 (Ahern et al. 2025): "We call a design an in silico success if (1) the root mean square deviation of all the heavy atoms in the catalytic residues is < 1.5 Å when aligned on the backbone N, C α , C of those catalytic residues in a Chai-1 prediction for at least one of the LigandMPNN sequences, and (2) the design contains no clashes with the ligand where a clash is defined as two atoms being within 1.5 Å." (bakerlab) (nature)

Translating the UMG-SP2 mechanism into a usable theozyme

The most practical recipe for a non-expert team is **not** to start from the published QM/MM transition state coordinates directly (those are in supplementary material of the Paiva and Świderek papers in Gaussian/CP2K-specific formats and require a chemist to extract). Instead:

1. Download the UMG-SP2 crystal structure from PDB (deposited alongside Li et al. *Adv. Sci.* 2025, ligand-free and ligand-bound co-structures at 2.59 / 2.16 / 2.40 Å resolution).
2. Extract the Ser190 / Ser166 / Lys91 side-chain atoms plus the bound short-chain carbamate substrate into a PDB.
3. Manually edit the carbamate to a transition-state mimic — specifically, sp³-hybridize the carbamate carbon and add a fourth substituent representing the attacking Ser-O γ (you build the *tetrahedral intermediate* analog, which is the standard TS approximation used in enzyme design). This step is what the Lauko (invrotzyme.py) script does for Ser-His hydrolases; for Ser-Ser-Lys you adapt the same logic.

4. Write a Rosetta `.params` file with `molfile_to_params.py` and a `.cst` file specifying the Ser-Oγ ... C(carbamate) distance (~2.5 Å), the Ser-Sercis H-bond, and the Lys-Nζ ... Ser-Sercis H-bond.

In practice, the simplest hackathon hack is to start from the existing `laukoag/serine-hydrolase-design` repo, which already has a Ser-His + tetrahedral-substrate constraint file, and **mutate the constraints to Ser-Ser-Lys** — copying the Lys91 position from the UMG-SP2 crystal as the third catalytic residue. This is the path that will work in 72 hours. [GitHub](#)

PART B — Tool-by-tool installability and runtime reality (as of early 2026)

RFdiffusion (original): [GitHub](#) `RosettaCommons/RFdiffusion`. BSD license, fully open. Docker image and conda env both work. Runs on a 16+ GB GPU (24 GB recommended for >200 residues). Install pain: moderate — `SE3Transformer` submodule is finicky on newer CUDA; use the official Apptainer/Singularity `.sif` if possible. Runtime: ~15 s/design (150 aa, A100) to ~10 min/design (small GPU). Hosted on Tamarind Bio with no install. [Nature](#)

RFdiffusionAA (All-Atom): [GitHub](#) `baker-laboratory/rf_diffusion_all_atom`. Open weights at `files.ipd.uw.edu`. Provides atom-level ligand awareness — the relevant tool if you want the diffusion model to "see" your urethane substrate. Apptainer-only install officially supported. ~30 s–2 min/design on A100. The only tool natively supporting ligand-aware backbone scaffolding before RFdiffusion2 was released. [GitHub](#) [LinkedIn](#)

RFdiffusion2: [GitHub](#) `RosettaCommons/RFdiffusion2`, released 15 Sept 2025; weights public. README warns it is "Under Construction — currently only inference is supported." Apptainer `.sif` provided; conda install possible but discouraged. Atomized motifs (e.g. enzyme active sites) take "up to 10 minutes" per case on an RTX 2060 and roughly 30 s–2 min/design on an A100. Critical for our use case because it accepts atom-level motifs without requiring rotamer enumeration. Hosted on Tamarind Bio as `RFdiffusion2`. [GitHub + 2](#)

RFdiffusion3: [GitHub](#) `RosettaCommons/foundry`, released 22 Dec 2025. Open weights and training code. ~10× faster than RFdiffusion2 per their reported benchmarks. Atom-level, unified across symmetric / binder / enzyme / nucleic-acid design tasks. The IPD provides a Colab tutorial running RFD3 + MPNN + RF3 end-to-end. Per the Foundry README, weights are managed via `foundry install base-models`. Hosted on Tamarind Bio as `RFdiffusion3`. **However:** RFdiffusion3 was released only ~6 months before the hackathon date and the community has had less time to write robust enzyme-design glue around it; for a 72-hour project the safer bet is RFdiffusion2 with the Lauko/Ahern recipe. [Uw](#) [GitHub](#)

ProteinMPNN / LigandMPNN / SolubleMPNN: [GitHub](#) `dauparas/LigandMPNN` (subsumes ProteinMPNN). MIT licensed. Trivial install (`pip install -r requirements.txt`). Runs on any GPU with >6 GB VRAM, often CPU-only is feasible. Sub-second per sequence per backbone. No friction.

AlphaFold2 / ColabFold: AF2 itself (DeepMind) is open-source Apache 2.0; ColabFold (Mirdita et al.) is MIT. ColabFold is the de facto choice — drop-in MMseqs2 MSA server replaces the slow Jackhmmer/HHblits step. Sub-minute per 300-residue protein on an A100; ~2–3 min on a 4090. Hosted free via Google Colab notebook. (Ranomics)

AlphaFold 3: GitHub (google-deepmind/alphafold3), public since Feb 2025. **Code is CC-BY-NC-SA 4.0 (non-commercial only). Weights are gated — must request and obtain explicit Google approval — and the WEIGHTS_TERMS_OF_USE.md classifies them as Google Confidential Information.** This makes AF3 a non-starter for a public hackathon demo, both because of license risk and because weights cannot be redistributed in a Docker image. **Use Boltz-2 instead** — it is MIT, fully open, ~20 s/structure on A100, and supports ligand binding affinity prediction natively. (GitHub)

ESM3 / ESM2 / ESM-C: GitHub (evolutionaryscale/esm). ESM2 is the original, fully open. **ESM3-open-small (1.4B params) is under the EvolutionaryScale Cambrian non-commercial license** — usable for hackathon prototyping but not for commercial use; ESM3 medium/large are API-only via Forge. ESM-C 300M is the only permissively-licensed (Cambrian Open) generative variant. For sequence-validation in a hackathon, **stick to ESM2 / ESMFold:** no license issues, fast single-sequence folding. (GitHub)

ESMFold: subset of ESM2, single-sequence structure prediction. Open. Useful as a fast complementary refold metric. ~10 s/sequence on 24 GB GPU.

Boltz-1 / Boltz-2: GitHub (jwohlwend/boltz). **MIT licensed for both code and weights, academic and commercial use both permitted.** Boltz-2 includes binding-affinity prediction explicitly. Runtime: ~20 s/structure (A100 — official MIT/Recursion figure), 1–5 min for ≤1300-token systems on an RTX 4090 with bfloat16 (UCSF ChimeraX testing). Install via pip. **This is the right structure-prediction tool for the hackathon.** NVIDIA NIM container needs ≥48 GB VRAM but the open-source pip build runs on 24 GB. (GitHub + 5)

Chai-1: GitHub (chaidiscovery/chai-lab). License permits academic use; recommended GPU is A100 80GB / H100 / L40S 48GB but "users have reported success with consumer-grade RTX 4090" for small complexes. Used internally by RFdiffusion2 as the structure-prediction filter. **Chai-2 is API-only and is antibody-focused** (zero-shot antibody design in a 24-well plate); not relevant to enzyme design. (GitHub) (GitHub)

PLACER: GitHub (baker-laboratory/PLACER). Weights bundled in repo. Conda env (placer_env.yml) provided. CUDA 12.1+, PyTorch 2.3, DGL 2.4. **Runtime: 1–3 s/model on GPU**, ~1 min/50-model ensemble (per README FAQ #12). Apple Silicon (MPS) supported via dedicated env. This is the single most important "active-site quality" tool and **must be in your pipeline** for credibility. (GitHub + 3)

RFpeptides / RFantibody: Subprotocols of RFdiffusion for macrocycles / antibodies. Not directly needed for urethanase design unless you pivot to peptide inhibitor design.

Rosetta / PyRosetta: Required for: (a) writing/parsing enzyme constraint files; (b) `molfile_to_params.py` for the substrate; (c) `FastRelax` cycles after LigandMPNN; (d) Rosetta `enzdes` for filter scores. **Free for academic use only; commercial requires license.** PyRosetta install via conda channel `conda.rosettacommons.org` and an academic license at `rosettacommons.org/software/license-and-download/`. Install painful but standard. For a hackathon, accept the academic license terms and proceed.

Supporting tools: RDKit (BSD, trivial install) for substrate cheminformatics, building a SMILES → 3D conformer for the carbamate. OpenMM (MIT) or GROMACS (LGPL) for any MD — **realistically out of scope for 72 h** beyond a token relaxation. AutoDock Vina or DiffDock can dock the substrate into the designed pocket, but Boltz-2 with ligand input already does this implicitly and is the better choice. **FoldSeek** (<https://search.foldseek.com/>) is the right tool for checking whether your final designs are novel folds — fast, hosted, gives you a "this fold has not been seen in nature" credibility line for the demo.

Hosted "no-DevOps" platforms (best to worst for this use case):

- **Tamarind Bio:** 100+ tools in one UI, including RFdiffusion, RFdiffusion2, RFdiffusion3, RFdiffusionAA, ProteinMPNN, LigandMPNN, SolubleMPNN, AlphaFold2, AlphaFold3, Boltz-1x, RFantibody, RoseTTAFold All-Atom. **Free tier is exactly 10 monthly jobs**, unlimited storage, email support, with an academic-upgrade form for university/non-profit emails. Python API. Premium pricing (per third-party reporting) starts ≈\$50K/year. **This is the recommended platform for the hackathon's final demoable runs.** `Y Combinator + 4`
- **NeuroSnap:** Similar offering — AlphaFold variants, RFdiffusion, RFdiffusion2, RoseTTAFold All-Atom, GROMACS, DiffAb, NeuroBind, DiffDock. Free tier offers 1 credit at signup ("Designed for academic use only, this plan includes 1 credit to start with, 2 parallel jobs"); per-tool credits charged at job submission. Good fallback to Tamarind. `Neurosnap`
- **ColabFold / ColabDesign (sokrypton):** Free Colab notebooks for ColabFold + RFdiffusion + ProteinMPNN. Excellent for quick exploration on a single GPU; not great for batch runs of thousands.
- **BioLib / Hugging Face Spaces:** Smaller hosted snippets, less comprehensive — useful for one-off tasks.

PART C — The realistic 72-hour execution path

Hour 0-8: setup and theozyme.

- Two members install Tamarind Bio + Boltz-2 locally; the other two clone `laukoag/serine-hydrolase-design`, `RosettaCommons/RFdiffusion2`, `baker-laboratory/PLACER`, and `dauparas/LigandMPNN` on a rented A100 instance (RunPod, Lambda, or Modal). Estimate \$20-80 of GPU time for the whole hackathon.

- Download UMG-SP2 crystal structure from PDB. Use PyMOL to extract Ser190, Ser166, Lys91, and the bound carbamate. Save as a stub PDB.
- Build the tetrahedral-intermediate analog of the carbamate-Ser190 complex in RDKit (sp³ carbamate C with Ser-Oy substituent). Save as both PDB and SDF/MOL2.
- Generate a Rosetta `params` file with `molfile_to_params.py` and a `cst` file with the three catalytic constraints. **This is the hardest step — budget a full team member on it with AI-coding-assistant help (Claude Code / Cursor reading the Rosetta cookbook and the Lauko repo simultaneously).**

Hour 8–32: backbone diffusion at scale.

- Use RFDiffusion2 to sample 1,000–5,000 backbones scaffolding the Ser-Sercis-Lys + carbamate motif. On a single A100 at ~30 s–2 min/design, that's ~12–24 h of GPU. Parallelize across 2–4 rented GPUs (Modal/RunPod) if budget allows; this is embarrassingly parallel.
- In parallel, run ColabDesign-style ProteinMPNN on a subset to sanity-check that backbones are designable. Save all to a structured directory tree (`outputs/{design_id}/backbone.pdb`).

Hour 32–56: sequence design + structure prediction.

- Run LigandMPNN with the substrate present, 4–8 sequences per backbone. ~1 second per sequence per backbone; trivial.
- Run Boltz-2 on every (sequence, ligand) pair. At ~20 s/structure on A100 and 4–8 sequences × 1000 backbones = 4,000–8,000 predictions = 22–44 h of GPU. **This is the time sink.** Mitigate by (a) filtering with ESMFold first (10 s/sequence) to cut down to top 25%, then (b) running Boltz-2 only on the survivors.
- Compute scRMSD between Boltz-2 prediction and original backbone; compute pLDDT, catalytic-residue RMSD.

Hour 56–68: PLACER preorganization filter + demo dashboard.

- Apply the Lauko 2025 cascade exactly: pLDDT > 75, global C α RMSD < 1.5 Å, catalytic C α RMSD < 1.0 Å. Expect ~5–20% survival, so ~200–1,600 designs remaining. (`bakerlab`)
- Run PLACER 50-model ensembles on the survivors (apo state only for speed). ~1 min/design. **Headline metric: fraction of designs where the Ser190-Lys91-Sercis catalytic H-bonds form in the great majority of 50 PLACER models** (Lauko 2025 used Ser-His H-bond formation in apo as the round-2 filter and reports that filter cut 98.4% of AF2-passing designs while raising hit-rate 3-fold).
- Build the React/FastAPI dashboard: protein viewer (NGL.js or Mol* embedded), per-design table sortable by pLDDT/scRMSD/PLACER-fraction, top 10 designs highlighted, FoldSeek "novel fold" badge for each.

Hour 68-72: polish, narrative, demo.

- Run FoldSeek on the top 10 designs to claim fold novelty.
- For each top design, produce a 3-panel visualization: (1) designed backbone with Ser/Ser/Lys highlighted, (2) Boltz-2 pose of the carbamate substrate, (3) PLACER ensemble overlay showing catalytic-geometry preservation.
- Write a 1-page methods note explaining each filter and citing Lauko 2025 / Ahern 2026 / Paiva 2025 / Świderek 2025.

What's genuinely achievable in 72 h (in-silico):

- Hundreds-thousands of de-novo backbones with a Ser-Sercis-Lys + carbamate-TS motif.
- Tens-hundreds of sequences passing the AF2/Boltz-2 self-consistency filter.
- A ranked top-10 with PLACER preorganization scores and FoldSeek novelty.
- A reproducible Docker/Modal pipeline + a React/FastAPI dashboard.

What is NOT achievable in 72 h:

- Any wet-lab expression, purification, or assay.
- Rigorous all-atom MD or QM/MM validation of the designs (each would take days per design).
- Quantitative kcat/Km predictions beyond Boltz-2's binding-affinity readout (which is a screening tool, not a turnover predictor).
- A confident claim that any specific design "is" a urethanase. The honest claim is "is computationally consistent with being a urethanase scaffold."

Demoable proxy metrics — what each means and what's "good"

Metric	Tool	Plain meaning	"Good" threshold (per literature)
pLDDT	AF2 / ColabFold / Boltz-2	Per-residue confidence in the predicted structure (0-100) Ranomics	>75 for serine hydrolase designs (Lauko 2025, exact value); >80 generally desirable; >90 is high confidence
scRMSD (self-consistency RMSD)	AF2 vs RFdiffusion backbone	Does the designed sequence fold back to the designed backbone?	<1.5 Å Cα RMSD globally (Lauko 2025, exact value); <2 Å is the common community threshold biorxiv
Catalytic-residue Cα RMSD	AF2 vs design	Are the catalytic residues in the right	<1.0 Å (Lauko 2025, exact value)

Metric	Tool	Plain meaning	"Good" threshold (per literature)
		place?	
PAE / PAE-interaction	AF2 multimer / Boltz-2	Pairwise alignment error, in Å — confidence in residue-residue distances (Ranomics)	<5 Å for confident contacts; community uses <10 Å for binder filtering (RFantibody)
PLACER ensemble preorganization	PLACER 50-model ensemble	Fraction of ensemble in which the catalytic H-bonds form	Stringent: filter cut 98.4% of AF2-passing designs in Lauko 2025 round 2 (1.6% survived), effectively requiring near-universal formation of the key catalytic H-bond across the 50-model ensemble
Boltz-2 binding-affinity readout	Boltz-2	A trained predictor of pKd/IC50 for the ligand in pocket	Boltz-2 achieves Pearson $r \approx 0.62$ on FEP+ benchmark (NVIDIA) (vs. FEP+ at $r \approx 0.72$), (biorxiv) at >1000× speed; relative ranking is more reliable than absolute values
RFdiffusion2 in-silico success	RFdiffusion2 + Chai-1 + LigandMPNN	Catalytic-heavy-atom RMSD <1.5 Å to a Chai-1 refold, no ligand clashes <1.5 Å (Ahern 2025, exact definition)	41/41 sites in AME benchmark satisfied this; (bioRxiv) the canonical "did the design work" check
FoldSeek novelty	FoldSeek vs PDB	TM-score to nearest natural fold	TM <0.5 = "no significant structural match" = novel fold (bakerlab)
Sequence recovery	ProteinMPNN	How similar are MPNN-designed sequences to the native at the same backbone	30–50% typical for de novo backbones; not directly a filter, more a sanity check
Rosetta cst_score	PyRosetta enzdes	Sum of constraint violations for the catalytic geometry	<2 Rosetta Energy Units (REU) per constraint; <1 REU is excellent

AI coding assistants and hosted platforms

Claude Code and Cursor are **highly effective** for this task because the bottlenecks are (a) writing

thin Python glue around well-documented tools, (b) parsing Rosetta config syntax, (c) writing a FastAPI/React dashboard. Empirically, an LLM coding assistant fed the `(serine-hydrolase-design)` repo + the `(LigandMPNN)` README + the Boltz-2 docs can produce working pipeline scripts in tens of minutes, vs the days it would take a non-expert to RTFM. The intellectual content — *which catalytic atoms go where, what tolerances are realistic* — is what an LLM cannot help with on its own; that's where the QM/MM literature on UMG-SP2 plus the Lauko Ser-His template comes in.

Tamarind Bio's main value-add is that it sidesteps

CUDA/PyTorch/DGL/openbabel/SE3Transformer dependency hell entirely, which is the most common 6–12-hour time sink in protein-design hackathons. The catch is the precise 10-job/month rate limit on the free tier; the realistic strategy is to use Tamarind for the *final* demo runs and a self-hosted A100 (Modal or RunPod, ~\$1–2/hr for A100 spot) for the bulk sampling.

PART D — Risk, shortcuts, contingencies

D.1 Biggest technical bottleneck: theozyme specification. Specifically, the chemistry of *writing a correct atom-level Rosetta `(params)` + `(cst)` description of a Ser-Sercis-Lys amidase tetrahedral intermediate with a urethane substrate*. The Lauko Ser-His template is close but not identical (the second residue is a Sercis, not a His; the substrate is a carbamate, not an ester). **De-risk by:** (a) literally copying the Ser190 / Ser166 / Lys91 coordinates and the bound carbamate from the UMG-SP2 crystal structure (Li et al. 2025), then (b) using RDKit/Open Babel to build the sp³ tetrahedral analog of the carbamate carbon, and (c) borrowing the Rosetta constraint syntax from the Lauko `(invrotzyme.py)` script with the constraint targets mutated.

D.2 Credibility-preserving shortcuts.

- Start from the published UMG-SP2 theozyme/crystal — do NOT try to design a novel theozyme. Cite Paiva 2025 and Świderek 2025 as the QM/MM basis.
- Use Boltz-2 instead of AlphaFold 3 (sidesteps the gated weights + non-commercial license).
- Use Tamarind Bio's hosted RFdiffusion2 + Protein Design pipeline for the demo. The judges can literally see the same dashboard the Baker lab uses.
- Use the Lauko `(laukoag/serine-hydrolase-design)` repo as a starting point — adapt the constraints, do not rewrite from scratch.
- Use benchmark substrates with known kinetic data on UMG-SP2: model dicarbamate substrates from Branson 2023 (Angew. Chem. DOI 10.1002/anie.202216220) and the 4-nitrophenyl benzylcarbamate (pNC) used in Świderek 2025. [\(Wiley Online Library\)](#)

D.3 MVP vs stretch artifact.

- **Minimum viable (everything goes wrong scenario):** A ProteinMPNN/LigandMPNN-only redesign of UMG-SP2 itself — keep the wild-type backbone, redesign all non-catalytic residues to improve predicted thermostability or solubility, refold with Boltz-2, compute

predicted-pKd shift for the substrate. This is a "scaffold optimization" demo and is achievable in 24 h by one person.

- **Stretch (everything goes right):** 5,000 RFdiffusion2 backbones → 20,000 LigandMPNN sequences → 2,000 Boltz-2 refolds → 200 PLACER-preorganized survivors → top-10 ranked + FoldSeek-novel + live React/FastAPI dashboard with substrate-pose visualization and downloadable PDB/FASTA for each. This is a competitive hackathon-winning artifact.

D.4 Honest assessment: Is it reachable?

Yes, but with a critical asterisk. A technically strong non-expert 4-person team CAN, in 72 hours, produce a *credible in-silico-only* de novo urethanase design artifact — credible in the sense that it follows the published Lauko/Ahern recipe, uses the same filter cascade with the same numeric thresholds, and produces designs that meet those thresholds. **They cannot honestly claim "we designed a working urethanase"** — that would require wet-lab validation, and even the Baker lab finds only 1-20% wet-lab hit rates on their best filter cascades (Lauko 2025 round 3: 18% 4MU-substrate hydrolysis among 132 designs; Ahern 2025 metallohydrolase: 11/96 functional). What they CAN honestly claim is: "We adapted the Baker lab's published serine-hydrolase de novo design pipeline (Lauko *Science* 2025, Ahern *Nature Methods* 2026) to a urethanase Ser-Sercis-Lys theozyme derived from the UMG-SP2 QM/MM mechanism (Paiva *Chem. Sci.* 2025, Świderek *JACS* 2025), generated N designs, filtered to top K by AF2/Boltz-2/PLACER metrics matching the published thresholds, and produced an open dashboard for browsing the results." (bakerlab) (nature)

The team will fail if: they try to build the theozyme from scratch from QM coordinates; they try to use AlphaFold 3 instead of Boltz-2; they try to run RFdiffusion locally on a 4090 for >1000 designs in 72 h; they skip PLACER (the headline credibility metric); or they spend >12 h on environment setup.

Adjacent reachable targets (if full de novo is too aggressive):

1. **Scaffold redesign:** keep the UMG-SP2 backbone, redesign non-catalytic surface with SolubleMPNN for thermostability + substrate-tunnel engineering with LigandMPNN. Lower risk, easier demo, but less novel.
 2. **Retrieval + screening pipeline:** FoldSeek/Foldcomp the entire UMG-SP2 against AlphaFold-DB / ESMAtlas to find natural amidase signature proteins, then ProteinMPNN-ddG to rank candidate mutations. This is a "wet-lab-prioritization" tool, not a design tool.
 3. **Binder design to UMG-SP2** instead of urethanase design — design a peptide/miniprotein that *binds and activates* UMG-SP2 at its allosteric L3 loop (Li et al. 2025 specifically calls out residues 219-226 as a regulatory loop). Uses RFdiffusion + ProteinMPNN with no enzyme-design chemistry needed. (ADS)
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Recommendations

1. **Commit to the de novo path with the Lauko/Ahern recipe**, but lock the theozyme to a copy of the UMG-SP2 crystal coordinates within hour 8. Do not entertain "let's improve the theozyme" debates after that point.
2. **Use Tamarind Bio for the demoable top-K runs and a self-hosted A100 (RunPod/Modal) for bulk sampling.** Budget \$20-80 in GPU time. Stockpile Tamarind's 10 free monthly jobs and request the academic upgrade with a university/non-profit email before the hackathon.
3. **Pipeline: RFdiffusion2 → LigandMPNN → ESMFold (cheap filter) → Boltz-2 (refold + binding affinity) → PLACER (preorganization).** Do NOT use AlphaFold 3. Boltz-2 is MIT, fast (≈ 20 s/structure on A100), ligand-aware, and is the right structure-prediction tool for this exact problem.
4. **Have one team member do the theozyme work full-time with AI-assistant pair-programming; the other three set up infra, the pipeline glue, and the dashboard.** Theozyme is the bottleneck, not infrastructure.
5. **Headline demo metric: "Fraction of designs in which the Ser190-Sercis-Lys91 catalytic geometry forms in the great majority of 50 PLACER apo-state models, restricted to designs with Boltz-2/AF2 pLDDT > 75, global C α RMSD < 1.5 Å, and catalytic-C α -RMSD < 1.0 Å (Lauko Science 2025 thresholds), and with FoldSeek TM-score < 0.5 to nearest natural amidase."** Quote this exactly to judges; it is defensible.
6. **Have a written contingency to the LigandMPNN-redesign-of-UMG-SP2 MVP** in case the theozyme step fails by hour 24.

Benchmarks that would change the plan

- If by hour 16 the team has not produced a single valid RFdiffusion2 backbone scaffolding the Ser-Sercis-Lys motif, pivot to the LigandMPNN-redesign MVP.
- If by hour 36 fewer than 20 designs pass AF2/Boltz-2 self-consistency (scRMSD < 1.5 Å, pLDDT > 75), broaden the substrate definition to a generic dicarbamate and re-run.
- If by hour 48 PLACER is not running locally, replace it with a Boltz-2 + Chai-1 ensemble (run 5 seeds, compute catalytic-residue RMSD variance) — less rigorous but still defensible.

Caveats

- **No published de novo urethanase exists** as of June 2026. The closest published work is metallohydrolase design (Kim, Woodbury, Ahern et al., *Nature* 649, 246-253, 2026, DOI 10.1038/s41586-025-09746-w, published online 3 Dec 2025) and serine hydrolase design (Lauko et al. *Science* 388, eadu2454, 18 April 2025) — both with low wet-lab hit rates (1-

20%). The hackathon cannot fairly claim a novel scientific result; it can claim a methodological adaptation. (Nature)

- **PLACER preorganization correlates with — but does not guarantee — wet-lab activity.** Lauko 2025 showed only 5.2% wet-lab hit rate even on PLACER-preorganized designs in round 2, rising to 18% (4MU hydrolysis) and 73% (within the 11 four-state-filtered designs that showed activity over background) only with the full multi-state filter cascade in round 3. In-silico preorganization is a necessary, not sufficient, condition.
- **Boltz-2 binding affinity is a screening tool, not a k_{cat}/K_m predictor.** It approaches FEP accuracy (Pearson $r \approx 0.62$ vs $r \approx 0.72$ for FEP+ on the CDK2/TYK2/JNK1/P38 benchmark) on equilibrium binding, but turnover requires modeling all four reaction states (apo, substrate, TI, AEI), which Boltz-2 cannot do directly. (NVIDIA) (biorxiv)
- **AlphaFold 3 weights are gated and non-commercial.** Do not use them in any artifact intended for redistribution, demoing in a Docker image, or for any commercial purpose. Boltz-2 is the right substitute.
- **ESM3 weights are non-commercial.** If used in any pipeline, the demo and code must be flagged as non-commercial.
- **The RFdiffusion2 README explicitly says "Under Construction"** as of its release; expect bugs, expect to file GitHub issues, expect to debug. The Lauko Ser-His pipeline (using RFdiffusion v1 + LigandMPNN) is more mature.
- **A "demoable artifact" is not a paper-ready result.** The right framing for judges is: "We produced a working, reproducible adaptation of the state-of-the-art de novo enzyme design pipeline to a new substrate class. We cannot claim catalysis; we claim computational consistency with the published Ser-Sercis-Lys mechanism."
- **The team's React/FastAPI/Docker/cloud strengths are exactly the right strengths.** The intellectual mismatch (no biochem expertise) is real but can be largely papered over by following the Lauko recipe literally and using a hosted platform — the pipeline does not require deep biochem to *run*, only to *invent*, and this hackathon does not require invention.